

How to produce bacI_panel_real.csv in Google Earth Engine

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A 12-step walkthrough. Total time: **≈30 min of clicking + 1–2 h of unattended export.**

What you will end up with

A single CSV file (bacI_panel_real.csv) with **192 rows × 9 columns** — the only file my pipeline needs to replace every synthetic number in the manuscript.

Before you start

You need:

1. A Google account with **Earth Engine** enabled (you already have one — pandasupranab@gmail.com).
2. Browser open at <https://code.earthengine.google.com>.
3. Your Cloud project set to **durable-pulsar-486209-b5** (top-right gear icon → “Project” → Select). This is your existing project.

You do **not** need:

- Python, conda, GEE client library, or any local install.
 - Any paid subscription.
 - Any institutional approval.
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Step 1 — Open the Earth Engine Code Editor

Go to <https://code.earthengine.google.com>. The screen has four panels:

- **Top-left:** script tabs (Scripts).
- **Top-centre:** the code editor (Editor).
- **Top-right:** Inspector / Console / **Tasks** tabs.
- **Bottom:** map.

If you see “Sign in”, sign in with pandasupranab@gmail.com.

Step 2 — Confirm the active Cloud project

Top-right of the page, click the **gear icon** — “Project”. Make sure the dropdown shows **durable-pulsar-486209-b5**. If it doesn’t, select it from the list.

Step 3 — Create a new script

In the top-left Scripts panel, click **NEW** → **File** → name it 04_phenology_extract. A blank editor opens in the centre.

Step 4 — Paste the script

Open the file I gave you, gee/04_phenology_extract.js, in any text editor (or open it from the GitHub repo at https://github.com/pandasupranab/RiceBaCI-GEE/blob/main/gee/04_phenology_extract.js). Select ALL contents and paste into the empty Code Editor.

Step 5 — Click Run

Top of the editor, click the blue **Run** button. After about 10–20 seconds you’ll see this message in the Console panel:

Module 04 ready. Open the Tasks tab → click Run on each of the 8 export tasks.

If you see a red error, screenshot it and send it to me — I’ll fix it on my side and reissue the script. The most common cause is a Cloud project mismatch (Step 2).

Step 6 — Open the Tasks tab

Top-right of the page, click the **Tasks** tab (between Inspector and Console). You should see **8 tasks** queued, named:

- bacI_panel_Baleshwar
- bacI_panel_Bhadrak
- bacI_panel_Kendrapara
- bacI_panel_Jagatsinghpur
- bacI_panel_Puri
- bacI_panel_Dhenkanal
- bacI_panel_Anugul
- bacI_panel_Cuttack

Each has a blue **Run** button next to it.

Step 7 — Click Run on every task

Click **Run** next to each of the 8 task names. A small dialog appears each time — just accept the defaults (the script has already set the destination to your Google Drive, folder `RiceBaCI_real_data`).

You can launch all 8 at once. They run in parallel on Google's servers — you do **not** need to keep the browser open while they run.

Step 8 — Wait for completion

Each task takes **5–15 minutes**. The Tasks panel shows progress (a spinning icon). When a task finishes, a green tick appears next to it.

You can close the tab and come back later. Tasks complete in the background.

Step 9 — Find the CSVs in Google Drive

Open Google Drive in another tab → navigate to **RiceBaCI_real_data** (auto-created in the root of your Drive). You should see 8 CSV files:

- `baCI_panel_Baleshwar.csv`
- `baCI_panel_Bhadrak.csv`
- ... (one per district)

Each contains 24 rows (8 years × 3 metrics) + 1 header row.

Step 10 — Concatenate the 8 CSVs into one

The easiest way:

1. Download all 8 CSVs to your laptop.
2. Open `baCI_panel_Baleshwar.csv` in Excel/LibreOffice.
3. Open the other 7 in turn, select rows 2–25 (data only, skip header), copy, paste at the bottom of the Baleshwar file.
4. Save As → name it **`baCI_panel_real.csv`**.

You should end up with **193 rows** (1 header + 192 data rows).

If you prefer one click, you can also just zip the 8 separate CSVs and send me the zip — I will concatenate.

Step 11 — Validate before sending

I've shipped a one-shot validator. Open Google Colab (or any Python environment), paste the file `scripts/validate_real_panel.py`, point it at your CSV. It checks:

- Exactly 192 rows
- All 9 columns present and correctly named
- District codes are the 8 expected ones
- Years are 2017–2024
- metric is one of SOS / POS / EOS
- value_days are between 1 and 366
- No nulls in mandatory columns

If validation passes, you'll see OK — panel ready for ingestion. If anything fails, the script tells you exactly which row.

Step 12 — Send the file

Drop `bacl_panel_real.csv` (or the zip of 8 CSVs) into a Google Drive folder, share the link with me, and reply here with the link.

I will then replace `synthetic_baci_panel.csv` in the repo, re-run the entire 13-stage pipeline, regenerate every figure and table with real numbers, and hand you the updated `Manuscript.docx` and `Supplement.docx`.

What if a value comes out wrong?

- Some districts in some years may show **null SOS** or **null EOS** if the rice crop never crossed the NDVI 0.4 threshold (heavily damaged year). That's fine — keep the row, leave the cell blank. My DiD pipeline handles missing cells.
 - If `n_pixels < 500` for a district-year, `qa_flag` will read excluded — also fine, the pipeline drops these from the main model and reports them separately in a robustness check.
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What if the GEE script errors

Most likely causes and fixes:

Symptom	Cause	Fix
"User memory limit exceeded"	Too many pixels in one district	Add <code>bestEffort: true</code> line is already there; if it still fails I'll switch to a tiled reducer

Symptom	Cause	Fix
"Image.normalizedDifference : Bands B8, B4 not found"	Wrong S2 collection ID	Confirm COPERNICUS/S2_SR_HARMONIZED (it is in the script)
"Cloud project not set"	Step 2 not done	Set project to durable-pulsar-486209-b5
Task stuck "READY" > 30 min	GEE backend busy	Cancel and re-run; usually resolves in ≤ 1 h

Just send me a screenshot of any error message and I'll patch it.